Experiment No. 7

Aim: Build an app to capture images and store in internal storage.

Apparatus / Software Required :

Hardware	- Laptop(ASUS Vivobook 16x) - Minimum 8 GB RAM - AMD Ryzen 5 5600H with Radeon Graphics Processor - Internet Connection
Software	- Windows 11 Operating System - Android Studio (Quail) - Java Development Kit (JDK) - Android SDK - Android Emulator (Pixel 7) - Gradle Build System

Theory

An Android app can obtain a photograph without implementing its own camera by launching the device's camera app with the MediaStore.ACTION_IMAGE_CAPTURE intent. With the Activity Result API this is done using ActivityResultContracts.TakePicture, which asks the camera app to write the full-resolution picture into a content Uri supplied by the caller. Because the camera app runs in a different process, the app cannot hand it a plain file path; a FileProvider is used to expose a file as a content:// Uri and temporarily grant write access.

Internal storage is the private directory of the application, returned by getFilesDir() (/data/data/<package>/files). Files stored there are not accessible to other apps, need no storage permission and are deleted when the app is uninstalled. Because the camera is launched through an intent, the CAMERA runtime permission is not required (it is required only if the app uses the Camera API / CameraX directly).

Large photos must be down-sampled with BitmapFactory.Options.inSampleSize before display to avoid OutOfMemoryError, and EXIF orientation must be applied so that the picture is shown upright.

Implementation Steps

Project Creation

- File > New > New Project > Empty Views Activity; Name: Camera Storage; Package: com.example.camerastorage; Language: Java; Min SDK 24.
- Add RecyclerView and ExifInterface dependencies in app/build.gradle and sync Gradle.

UI Design

- activity_main.xml: Capture button, ImageView for preview, TextViews for path/hint/empty message and a RecyclerView (grid).
- item_image.xml: a single ImageView used for each thumbnail.

Functionality

- Declare the FileProvider in the manifest and map internal 'images/' folder in res/xml/file_paths.xml.
- On button click create a file IMG_<timestamp>.jpg in getFilesDir()/images and get its Uri through FileProvider.getUriForFile().
- Launch the camera using the TakePicture contract; in the result callback show the preview or delete the empty file if cancelled.
- Load all .jpg files from the folder into the RecyclerView (newest first).
- Click a thumbnail to preview; long-press to delete after confirmation.

Running the App

- Run on a device or emulator. On the emulator set the camera to Virtual Scene or Webcam in the AVD advanced settings.
- Tap Capture Photo, take the picture and confirm. The image appears in the grid and the saved path is shown.
- Optionally open Device Explorer (View > Tool Windows > Device Explorer) > data/data/com.example.camerastorage/files/images to verify the files.

Files Used / Code Changes

File	Purpose / Change
MainActivity.java	Camera launch, file creation, result handling, list loading, preview and delete.
ImageAdapter.java	RecyclerView adapter showing saved images as thumbnails.
ImageUtils.java	Loads down-sampled, EXIF-rotated bitmaps.
activity_main.xml, item_image.xml	Main screen layout and grid item layout.
res/xml/file_paths.xml	FileProvider path configuration for internal storage.
AndroidManifest.xml	FileProvider, camera feature (optional) and queries element.
strings.xml, colors.xml, themes.xml	Text, colours and theme.
app/build.gradle, settings.gradle, build.gradle	Gradle configuration and dependencies.

Components Used

Component	Use in this project
Activity (AppCompatActivity)	Main screen.
Implicit Intent (via TakePicture	Launches the camera app.

contract)	
FileProvider	Shares an internal file as a content Uri with the camera app.
RecyclerView + GridLayoutManager + Adapter	Thumbnail grid.
ImageView, Button, TextView	UI widgets.
AlertDialog	Delete confirmation.
Internal storage (getFilesDir)	Private location where images are saved.

Important XML Attributes Used

Attribute	Meaning
provider android:authorities="${applicationId}.fileprovider"	Unique authority of the FileProvider.
android:grantUriPermissions="true"	Allows temporary write access to the camera app.
android:exported="false"	Provider is not public.
android.support.FILE_PROVIDER_PATHS	Points to the XML file listing shareable folders.
<files-path name path>	Maps getFilesDir() sub-folder to the provider.
uses-feature android.hardware.camera required=false	Installable without a camera.
<queries><intent><action IMAGE_CAPTURE/>	Package visibility for camera apps (Android 11+).
android:scaleType="fitCenter / centerCrop"	How images fit the ImageView.

Important Java Methods Used

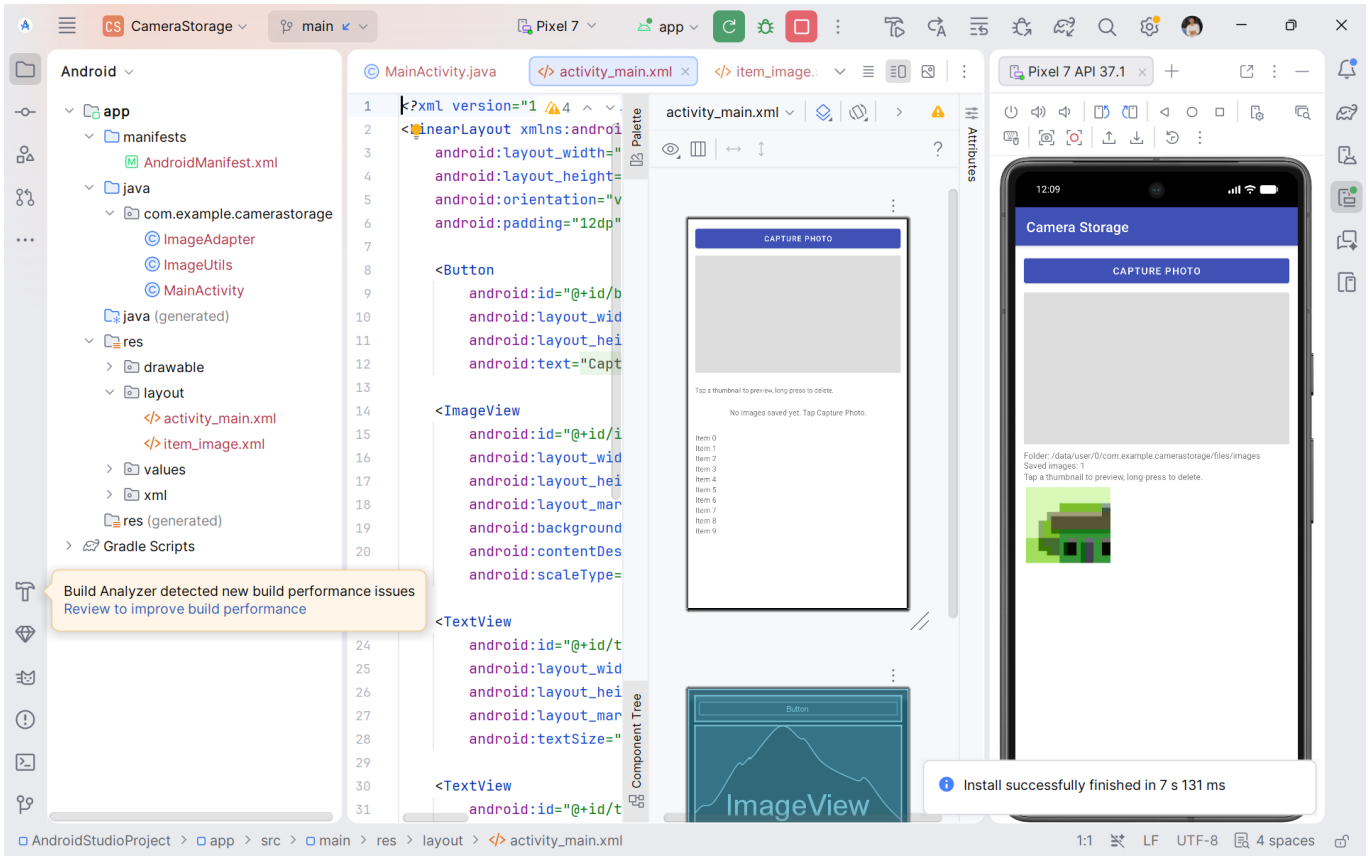
Method / Class	Purpose
registerForActivityResult(new TakePicture(), callback)	Registers camera launcher and result callback.
launch(Uri)	Starts the camera app.
FileProvider.getUriForFile()	Creates content Uri for a File.
getFilesDir()	Returns internal storage directory.
File.createNewFile(), delete(), listFiles(), length()	File handling operations.
BitmapFactory.decodeFile(), inSampleSize	Loads and down-samples images.
ExifInterface.getAttributeInt()	Reads image orientation.
Matrix.postRotate(), Bitmap.createBitmap()	Rotates the bitmap.
RecyclerView.Adapter methods	onCreateViewHolder, onBindViewHolder, getItemCount.
onSaveInstanceState()	Preserves pending file path on rotation.

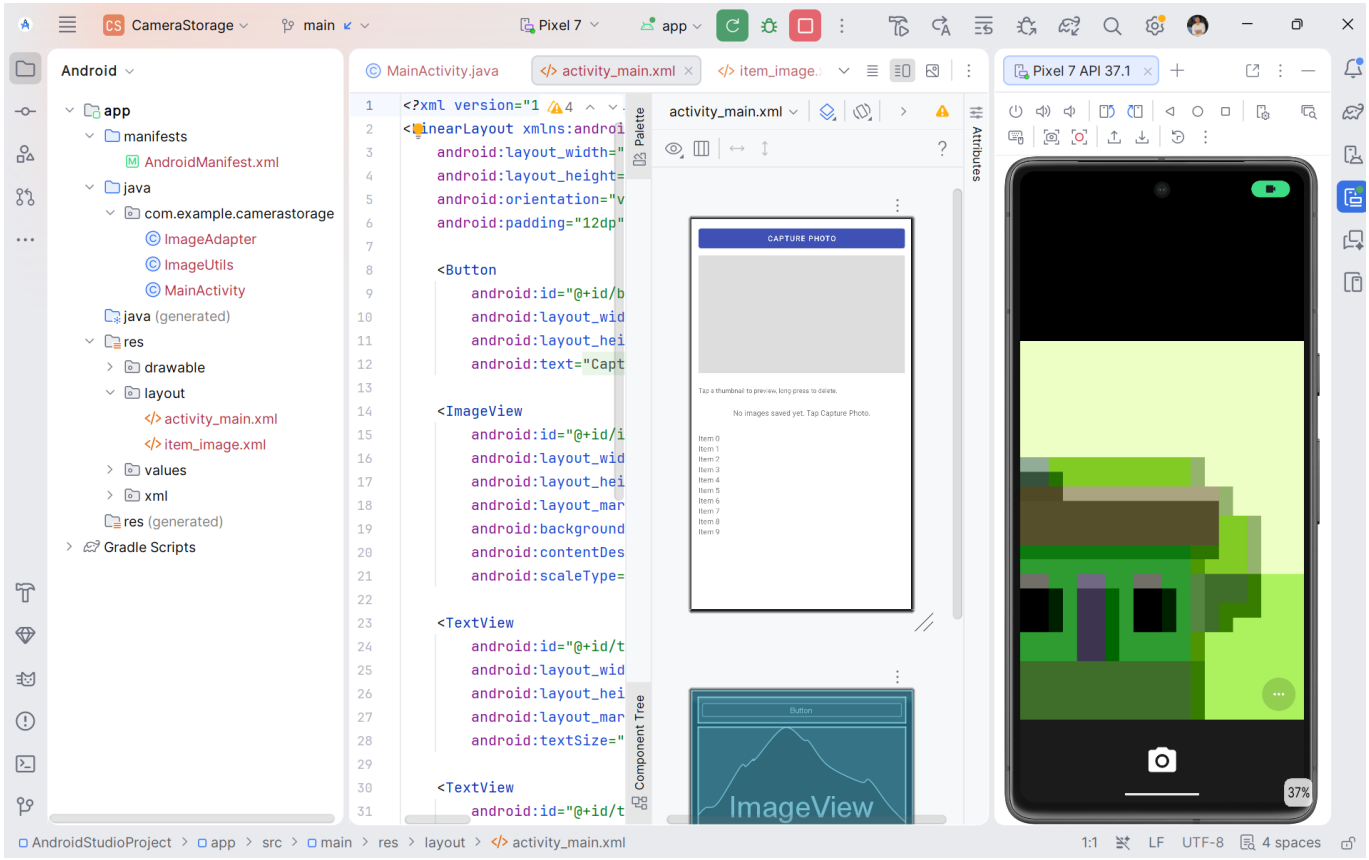
10. Errors Faced & Troubleshooting

The following are common errors for this experiment and their fixes. Add any additional errors you personally encounter while running it.

Error / Problem	Cause and Fix
IllegalArgumentException: Failed to find configured root	file_paths.xml does not contain files-path for 'images/'. Check name/path.
ActivityNotFoundException / no camera	Device or emulator has no camera app. Configure AVD camera or use a real phone.
Image not appearing after capture	Camera cancelled or Uri authority mismatch. Authority must equal getPackageName() + ".fileprovider".
OutOfMemoryError when showing photo	Bitmap decoded at full size; use inSampleSize (ImageUtils).
Photo displayed sideways	EXIF rotation not applied; use ExifInterface as in ImageUtils.
SecurityException: Permission Denial	Missing grantUriPermissions=true or provider not declared in manifest.

Output





Conclusion

The experiment shows how to capture a photograph through the camera app, save it securely in the app's internal storage using FileProvider, and display, list and delete the stored images. The project is complete as source code; run it on a device or emulator with a camera to obtain real output screenshots.

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